

Lab 3: Setting Up MySQL Database on your GCP

There 3 different ways to install MySQL database on your GCP Compute Engine as below:

1. Google Cloud SQL
2. Google Cloud Launcher
3. MySQL on Compute Engine

In this exercise lab, we are going to use Option 1 to create MySQL instance. However, students can create MySQL instance using either Option 2 or 3.

Before you start Lab 3, you should download exercise lab files from Khoury Github:

```
$ git clone https://github.khoury.northeastern.edu/jelee0408/CS5200_Lab.git
```

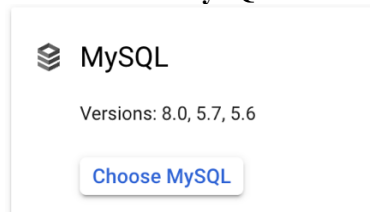
Check if all lab files are downloaded correctly.

A. Creating MySQL instance

1. Go to the Cloud SQL Instances page in the Google Cloud Platform Console.

GCP Console -> SQL

2. Then, **click “Create Instance”**
3. Select **MySQL** and click **Next**.
4. Click **Choose MySQL**.



5. Chose a Cloud SQL edition: **“Enterprise”**
6. Choose a preset for this edition: **“Sandbox”**
7. Enter the following info:
 - Instance ID: **neu-test-db**
 - Root password:
 - Choose database version: MySQL 5.6 or **MySQL 5.7** (default)
 - Choose a configuration to start with: **Development**
 - Region: **us-west1 (Oregon)**
 - Zone availability: **Single zone**
 - Primary zone: Any
8. Show “Customize your instance” option to change any of the following options:
 - Machine Type: **Shared core (1 vCPU, 0.614 GB)**

- Storage type: **HDD**
- Storage capacity: **10 GB**
- **Enable automatic storage increase**
- Set connectivity: check **Public IP**
- New network:
 - Name: **All IP**
 - Network: **0.0.0.0/0**
 - Then, click “Done”
- Backups: **uncheck automatic backups** and **Instance deletion protection**

9. Click **Create** (It will take approximately 10 ~ 15 minutes)

Pricing estimate

\$0.01 per hour (estimated, without discounts)

That's about \$0.28 per day.

Feature usage and traffic costs aren't included in estimate

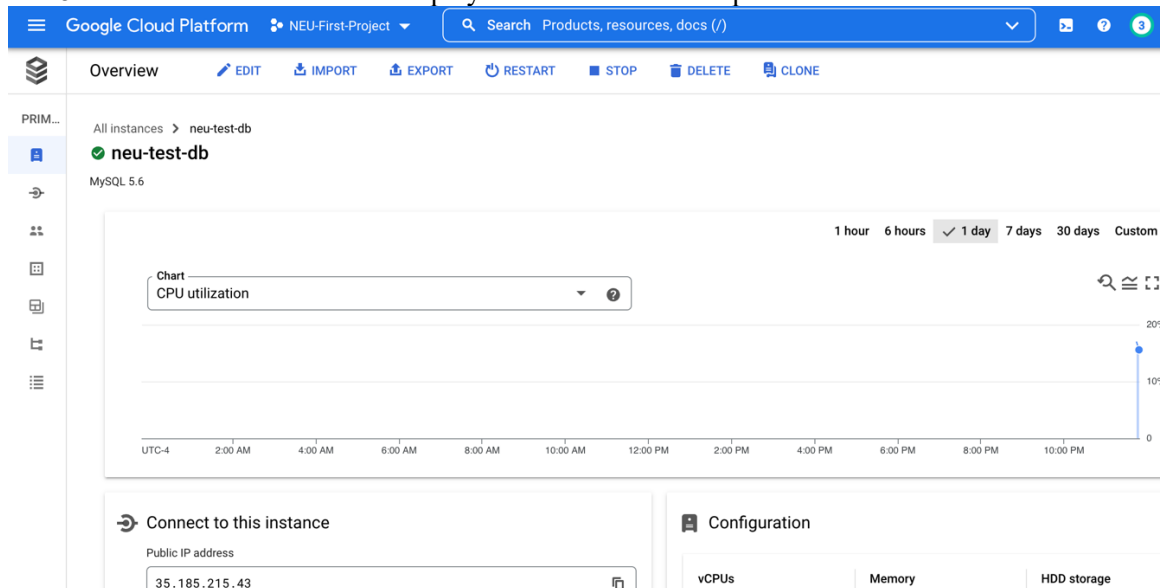
[▼ SHOW COST BREAKDOWN](#)

Summary

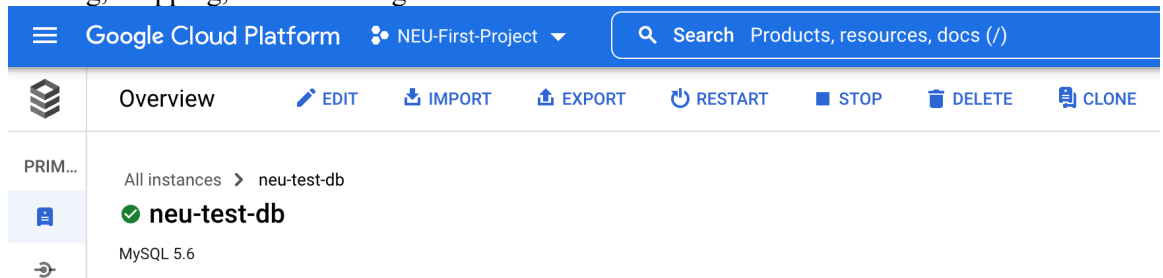
Cloud SQL Edition 	Enterprise
Region	us-west1 (Oregon)
DB Version	MySQL 5.7
vCPUs	1 vCPU
Memory	628.74 MB
Data Cache	Disabled
Storage	10 GB
Connections	Public IP
Backup	Manual
Availability	Single zone
Point-in-time recovery	Disabled
Network throughput (MB/s) 	125 of 125
Disk throughput (MB/s) 	Read: 1.2 of 125.0 Write: 1.2 of 107.8
IOPS 	Read: 7.5 of 1,000 Write: 15 of 10,000

B. Maintaining MySQL instance

1. Viewing information about MySQL instance
 - Select an instance to view by clicking the instance name.
 - Cloud Platform Console displays the Instance details pane for the instance.



2. Starting, Stopping, and Restarting Instances



- Starting an Instance
 - Select an instance to start by clicking the instance name.
 - Click Start, in the button bar at the top of the page.
 - In the Start database instance? dialog box, click Start.
 - The activation policy of the instance is set to Always and the instance is started.
- Stopping an Instance
 - Select an instance to restart by clicking the instance name.
 - Click Stop, in the button bar at the top of the page.
 - In the Stop database instance? dialog box, click Stop.
 - The activation policy of the instance is set to Off (Never) and the instance is stopped.
- Restarting an Instance
 - Select an instance to restart by clicking the instance name.
 - Click Restart, in the button bar at the top of the page.
 - In the Restart database instance dialog box, click Restart.

- Deleting an Instance
 - Select an instance to delete by clicking the instance name.
 - Click Delete.
 - In the Delete database instance dialog box, click OK.

C. Connecting to MySQL instance

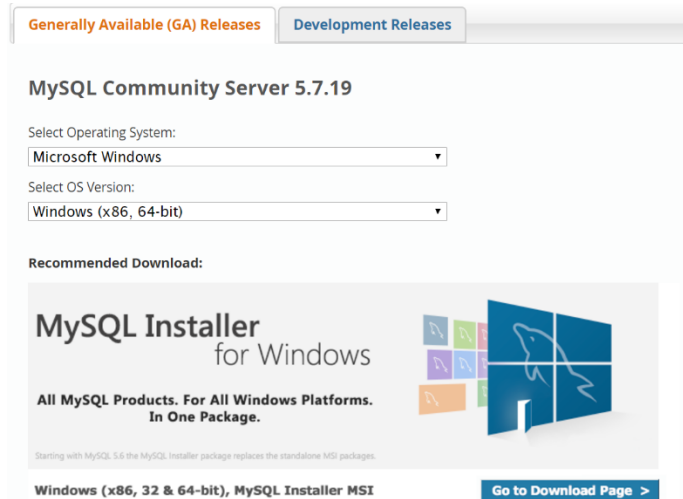
There are many different ways to connect MySQL instance running on GCP that you created before. Among them, you are going to test the following methods:

1. Connecting MySQL instance from your PC using IP Addresses
2. Connecting MySQL Client from Compute Engine
3. Connecting from App Engine
4. Connecting to Cloud SQL from External Applications

We will practice for the first method here, but you can use other connecting methods using the following instructions:

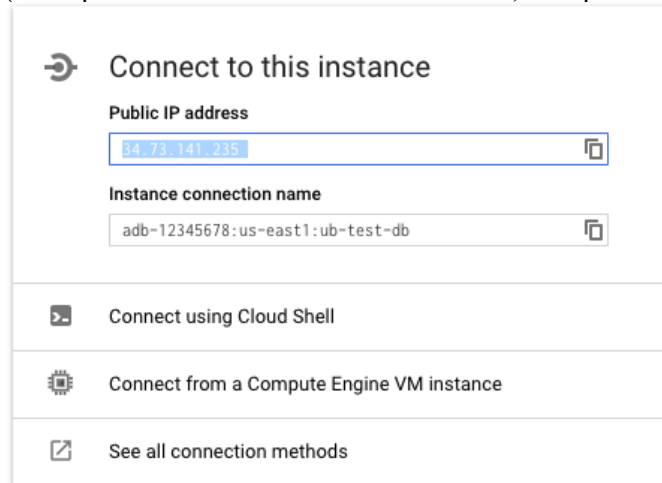
<https://cloud.google.com/sql/docs/mysql/how-to>

1. Connecting MySQL instance from your PC using IP Addresses
 - Install the mysql client on your PC (e.g., Windows or Mac-OS)
 - 1) Going to MySQL Community Server download page.
<https://dev.mysql.com/downloads/mysql/>
 If you want to install MySQL client only
<https://dev.mysql.com/downloads/shell/>
 - 2) Download the files



- 3) Install the Community Server (or extract zip file where you want), following the directions on the download page.
- Configure access to your Cloud SQL instance
 - 1) Click the instance to open its Overview page, and record its IPv4 address.
 - MySQL instance IP address: **11.11.11.11**

(Your public IP address should be different, so replace 11.11.11.11 by yours)

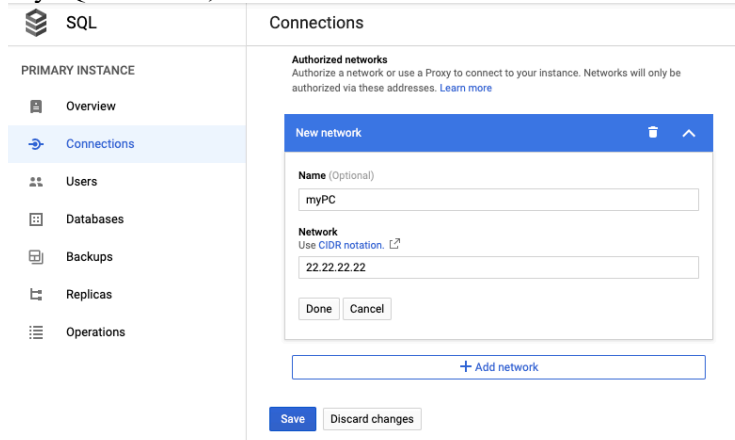


Skip 2) ~ 4) if you already created “All IP” network when you created an instance.

- 2) Log in to the client machine where your mysql client (i.e., your PC IP address) is installed to get external IP address
If you are not sure how to get it, click <http://ipv4.whatismyv6.com/>
 - Client IP address: 22.22.22.22

Note: You need to update every time you connect to mysql, as it changes.

- 3) MySQL instance, and **Connection** menu. Select the **Authorized Network**.



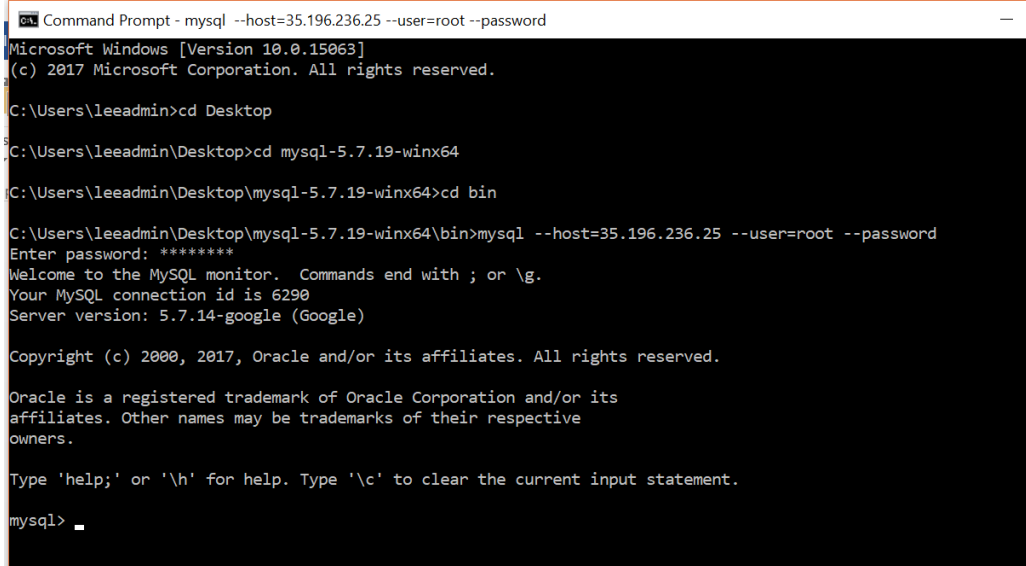
- 4) Click **Done**, then click **Save** at the bottom of your page to save your changes

- Connect to your Cloud SQL instance
 - 1) Windows:
 - Start cmd.exe on Windows
 - Go to where you extract mysql downloaded file on
 - **C:/Program Files/MySQL/MySQL Workbench 6.3 CE – This varies based on your path selection during installation**
 - 2) MacOS:
 - Start Terminal on MacOS

- 3) Start the mysql client with IP address for MySQL instance

```
mysql --host=11.11.11.11 --user=root --password
```

- 4) Enter your password
- 5) You should see mysql prompt



```
Command Prompt - mysql --host=35.196.236.25 --user=root --password
Microsoft Windows [Version 10.0.15063]
(c) 2017 Microsoft Corporation. All rights reserved.

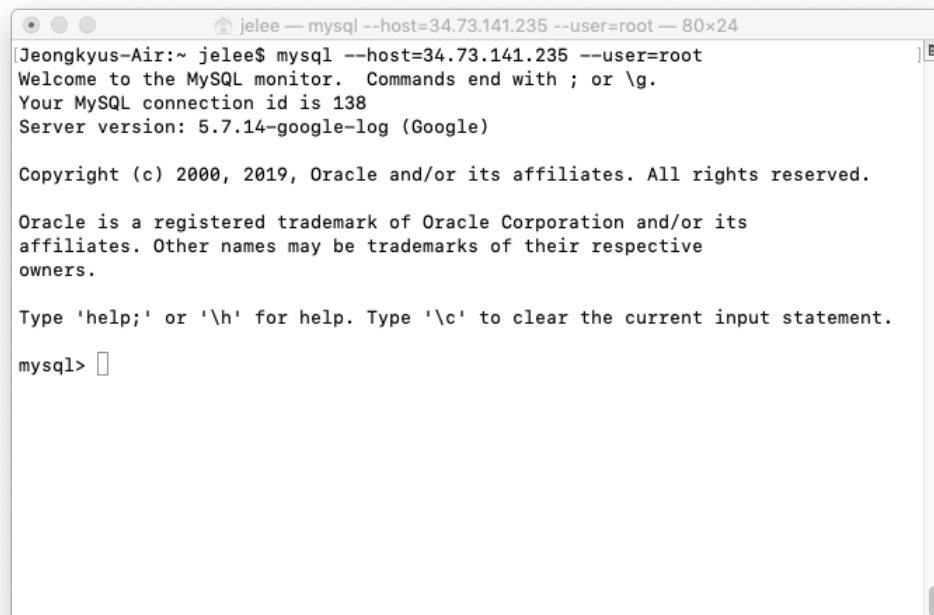
C:\Users\leeadmin>cd Desktop
C:\Users\leeadmin\Desktop>cd mysql-5.7.19-winx64
C:\Users\leeadmin\Desktop\mysql-5.7.19-winx64>cd bin
C:\Users\leeadmin\Desktop\mysql-5.7.19-winx64\bin>mysql --host=35.196.236.25 --user=root --password
Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 6290
Server version: 5.7.14-google (Google)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```



```
jelee — mysql --host=34.73.141.235 --user=root — 80x24
Jeongkyus-Air:~ jelee$ mysql --host=34.73.141.235 --user=root
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 138
Server version: 5.7.14-google-log (Google)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql>
```

For more information about GCP: <https://goo.gl/NYDpBc>

Exercise

After performing Section C, do the following and then submit the captured file to Canvas.

- Capture a screenshot of mysql that shows success of creating a sample table.
- **[Optional]** Capture screenshot of mysql that can connect MySQL instance using the following methods:
 - Connecting MySQL Client from Compute Engine
 - Connecting from App Engine
- Submit the screen shot into Canvas. When you submit it, you should indicate your name and NEU ID